Publication List

Articles

[1] B. Hertzsch, **J. Feldbrugge**, and R. van de Weygaert, "Galaxy populations in the IllustrisTNG caustic skeleton," *Monthly Notices of the Royal Astronomical Society* 552, stag1724 (2026). arXiv:2604.18209 [astro-ph.CO]

[2] S. Crew and **J. Feldbrugge**, "Asymptotic evaluation of multidimensional oscillatory integrals" (2026). arXiv:2610.22767 [math-ph]

[3] **J. Feldbrugge**, S. Crew, and U.-L. Pen, "Resurgence and hyperasymptotics in wave optics astronomy," *Journal of Cosmology and Astroparticle Physics* 2026(09), 022 (2026). arXiv:2602.21493 [astro-ph.CO]

[4] **J. Feldbrugge** and U.-L. Pen, "The real-time Feynman path integral for step potentials," *Annals of Physics* 490, 170490 (2026). arXiv:2508.17578 [quant-ph]

[5] A. Weber, **J. Feldbrugge**, and E. Pisanty, "Universal approach to saddle-point methods in attosecond science," *Physical Review A* 113, 063111 (2026). arXiv:2510.12545 [quant-ph]

[6] I. Aniceto, **J. Feldbrugge**, and C. J. Howls, "Which saddles contribute? The southeast rule for multidimensional integrals" (2026). arXiv:2606.28271 [math-ph]

[7] A. Read, **J. Feldbrugge**, C. Boehm, R. van de Weygaert, and B. Hertzsch, "Caustic skeleton and the local cosmic web: the Coma cluster node and the Pisces-Perseus ridge" (2026). arXiv:2604.22213 [astro-ph.CO]

[8] B. Hertzsch, **J. Feldbrugge**, M. Rodriguez, and R. van de Weygaert, "A new recipe for caustic pancakes: on the reality of walls in the cosmic web," *Journal of Cosmology and Astroparticle Physics* 2026(02), 037 (2026). arXiv:2510.02419 [astro-ph.CO]

[9] **J. Feldbrugge** and R. van de Weygaert, "What makes a cosmic filament? The dynamical origin and identity of filaments – I. Fundamentals in 2D," *Monthly Notices of the Royal Astronomical Society* 539, 873–901 (2025). arXiv:2405.20475 [astro-ph.CO]

[10] **J. Feldbrugge**, D. L. Jow, and U.-L. Pen, "Complex classical paths in quantum reflections and tunneling," *Physical Review D* 111, 085027 (2025). arXiv:2309.12420 [quant-ph]

[11] **J. Feldbrugge** and J. Y. L. Jones, "Efficient evaluation of real-time path integrals," *Physical Review D* 111, 083524 (2025). arXiv:2501.16323 [quant-ph]

[12] B. Bonga, **J. Feldbrugge**, and A. Metidieri, "Wave optics for rotating stars," *Physical Review D* 111, 063061 (2025). arXiv:2410.03828 [gr-qc]

[13] **J. Feldbrugge**, "Phase-space Delaunay tessellation field estimator," *Monthly Notices of the Royal Astronomical Society* 536, 807–815 (2025). arXiv:2402.16234 [astro-ph.CO]

[14] **J. Feldbrugge**, Y. Yan, and R. van de Weygaert, "Statistics of tidal and deformation eigenvalue fields in the primordial Gaussian matter distribution: the two-dimensional case," *Monthly Notices of the Royal Astronomical Society* 526, 5031–5053 (2023). arXiv:2301.07200 [astro-ph.CO]

[15] **J. Feldbrugge**, "Complex evaluation of angular power spectra: Going beyond the Limber approximation," *Physical Review D* 108, 103007 (2023). arXiv:2304.13064 [astro-ph.CO]

[16] D. L. Jow, U.-L. Pen, and **J. Feldbrugge**, "Regimes in astrophysical lensing: refractive optics, diffractive optics, and the Fresnel scale," *Monthly Notices of the Royal Astronomical Society* 525, 2107–2124 (2023). arXiv:2204.12004 [astro-ph.HE]

[17] **J. Feldbrugge**, D. L. Jow, and U.-L. Pen, "Crossing singularities in the saddle point approximation" (2023). arXiv:2309.12427 [quant-ph]

[18] **J. Feldbrugge** and N. M. D. Niezink, "Orthogonality relations for conical functions of imaginary order" (2023). arXiv:2309.05616 [math.CA]

[19] **J. Feldbrugge** and N. Turok, "Existence of real time quantum path integrals," *Annals of Physics* 454, 169315 (2023). arXiv:2207.12798 [hep-th]

[20] **J. Feldbrugge**, "Multiplane lensing in wave optics," *Monthly Notices of the Royal Astronomical Society* 520, 2995–3006 (2023). arXiv:2010.03089 [astro-ph.CO]

[21] **J. Feldbrugge**, U.-L. Pen, and N. Turok, "Oscillatory path integrals for radio astronomy," *Annals of Physics* 451, 169255 (2023). arXiv:1909.04632 [astro-ph.HE]

[22] **J. Feldbrugge** and R. van de Weygaert, "Cosmic web & caustic skeleton: non-linear constrained realizations – 2D case studies," *Journal of Cosmology and Astroparticle Physics* 2023(02), 058 (2023). arXiv:2212.07840 [astro-ph.CO]

[23] G. Wilding, K. Nevenzeel, R. van de Weygaert, G. Vegter, P. Pranav, B. J. T. Jones, K. Efstathiou, and **J. Feldbrugge**, "Persistent homology of the cosmic web – I. Hierarchical topology in ΛCDM cosmologies," *Monthly Notices of the Royal Astronomical Society* 507, 2968–2990 (2021). arXiv:2011.12851 [astro-ph.CO]

[24] **J. Feldbrugge** and N. Turok, "Gravitational lensing of binary systems in wave optics" (2020). arXiv:2008.01154 [gr-qc]

[25] A. Di Tucci, **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "Quantum incompleteness of inflation," *Physical Review D* 100, 063517 (2019). arXiv:1906.09007 [hep-th]

[26] **J. Feldbrugge**, M. van Engelen, R. van de Weygaert, P. Pranav, and G. Vegter, "Stochastic homology of Gaussian vs. non-Gaussian random fields: graphs towards Betti numbers and persistence diagrams," *Journal of Cosmology and Astroparticle Physics* 2019(09), 052 (2019). arXiv:1908.01619 [astro-ph.CO]

[27] P. Pranav, R. van de Weygaert, G. Vegter, B. J. T. Jones, R. J. Adler, **J. Feldbrugge**, C. Park, T. Buchert, and M. Kerber, "Topology and geometry of Gaussian random fields

I: on Betti numbers, Euler characteristic, and Minkowski functionals," *Monthly Notices of the Royal Astronomical Society* 485, 4167–4208 (2019). arXiv:1812.07310 [astro-ph.CO]

[28] **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "Inconsistencies of the new no-boundary proposal," *Universe* 4, 100 (2018). arXiv:1805.01609 [hep-th]

[29] **J. Feldbrugge**, R. van de Weygaert, J. Hidding, and Joost Feldbrugge, "Caustic skeleton & cosmic web," *Journal of Cosmology and Astroparticle Physics* 2018(05), 027 (2018). arXiv:1703.09598 [astro-ph.CO]

[30] **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "No rescue for the no boundary proposal: Pointers to the future of quantum cosmology," *Physical Review D* 97, 023509 (2018). arXiv:1708.05104 [hep-th]

[31] **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "No smooth beginning for spacetime," *Physical Review Letters* 119, 171301 (2017). arXiv:1705.00192 [hep-th]

[32] **J. Feldbrugge**, J.-L. Lehners, and N. Turok, "Lorentzian quantum cosmology," *Physical Review D* 95, 103508 (2017). arXiv:1703.02076 [hep-th]

[33] **J. Feldbrugge**, J. Hidding, and R. van de Weygaert, "Statistics of caustics in large-scale structure formation," *The Zeldovich Universe: Genesis and Growth of the Cosmic Web*, IAU Symposium 308, 107–114 (2016). arXiv:1412.5121 [astro-ph.CO]

[34] R. van de Weygaert, G. Vegter, H. Edelsbrunner, B. J. T. Jones, P. Pranav, C. Park, W. A. Hellwing, B. Eldering, N. Kruithof, E. G. P. Bos, J. Hidding, **J. Feldbrugge**, E. ten Have, M. van Engelen, M. Caroli, and M. Teillaud, "Alpha, Betti and the megaparsec universe: on the topology of the cosmic web," *Transactions on Computational Science XIV*, Lecture Notes in Computer Science 6970, 60–101 (2011). arXiv:1306.3640 [astro-ph.CO]

Theses

[35] **J. Feldbrugge**, "Path integrals in the sky: Classical and quantum problems with minimal assumptions," PhD thesis, Perimeter Institute, University of Waterloo, supervised by N. Turok (2019). Available online.

[36] **J. Feldbrugge**, "Primordial non-Gaussianity and large-scale structure," Part III Essay, University of Cambridge, supervised by P. Shellard and T. Giannantonio (2015). Available online.

[37] **J. Feldbrugge**, "Statistics of caustics in large-scale structure formation," Master thesis, University of Groningen, supervised by R. van de Weygaert, D. Roest, A.E. van Enter (2014). Available online.

[38] **J. Feldbrugge** and M. van Engelen, "Analysis of Betti numbers and persistence diagrams of two-dimensional Gaussian random fields," Bachelor thesis, University of Groningen, supervised by R. van de Weygaert, E. Pallante, G. Vegter (2012). Available online.